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Google Scholar: [↔ Published Work](#)

Summary

- Senior Applied Researcher for GitHub Copilot CLI.
Prev: Research Scientist at Microsoft Research.
- Publications in NLP/AI venues, including the *Best Paper Award* at AACL 2023.
- Areas of Expertise:
 - ◊ Designing agent harnesses for software engineering.
 - ◊ Training & contextualizing LLMs for code generation.
 - ◊ Mitigating factual hallucination in agents with code verification and knowledge base management.

Education

The University of Edinburgh

Ph.D. Informatics 2023

Thesis: *Inference of Natural Language
Predicates in the Open Domain*

Advisor: Mark Steedman

M.Sc. Artificial Intelligence, *Distinction* 2019

Brown University

B.Sc. Computer Science 2017

Professional Experience

GitHub Applied Science

Senior Applied Researcher

Sep 2026 → Present

Applied Researcher III

Apr 2025 → Aug 2026

Leading projects in multi-agent planning, verification, and orchestration in Copilot CLI agent harness.

- Researched and built Rubber Duck critique system, the first model-ensemble feature for Copilot CLI (combines Claude and GPT). Closed the Sonnet-Opus accuracy gap by 74.7% at only 5.5% avg cost to users, and increased user retention by 3.2%. Featured @ Microsoft Build 2026 keynote.
- Researched and built `update_todo`, the first tool for Copilot CLI to make complex plans. This improved resolution rate on long, multi-step problems by 5.8% without increasing total step count.
- Improved the architecture of fleet mode (subagent parallelization leveraging todos) for Copilot CLI, increasing resolution rate by 5% and decreasing wall clock time by 4.4% for complex tasks.

Microsoft Research

Research Scientist

Feb 2024 → Apr 2025

Member of the Excel AI research team.

- Led project to finetune code-generating LLMs for low-resource programming languages (e.g. Excel Formulas) and APIs, useful for integrating code generating LLMs with native applications.
- Developed multi-GPU training harness and novel synthetic data generation techniques.

Amazon Alexa AI

Applied Scientist II Intern

Jun 2022 → Nov 2022

- Researched and published a prototype for Alexa’s trivia question-answering. I improved a bespoke neural QA architecture to support knowledge graph expansion free of retraining cost.

Skills

- **AI & Machine Learning:** Designing multi-agent systems with tools for computer access; Mitigating factual hallucination in LLMs by generating executable code, and grounding to knowledge; Training LLMs in multi-GPU environments; Designing synthetic data for model finetuning; Statistical analysis.
- **Languages:** Python, TypeScript, Java, Bash, SQL/KQL
- **Tools:** PyTorch, Huggingface Transformers, LLMs like GPT-5, Qwen, and Claude, Git, Numpy, Scipy

Awards, Recognition, and Patents

Microsoft Build 2026 keynote demo for my feature Rubber Duck on Copilot CLI	2026
US Patent (Pending) “Generating Interactions for Question Answering on Structured Data”	2025
The <i>Best Paper Award</i> at AACL 2023: <i>Smoothing Entailment Graphs with Language Models</i>	2023
Huawei Ph.D. Scholarship Award	2019
Outstanding M.Sc. Dissertation: <i>Learning Negation Scope Semantics with Structure</i>	2019

Research Publications

LREC 2026	Nick McKenna , Xinnuo Xu, Jack Williams, Nicholas Wilson, Benjamin Van Durme, Christian Poelitz. <i>Synthetic Function Demonstrations Improve Generation in Low-Resource Programming Languages</i> .	↪ Link
LREC 2026	Christian Poelitz and Nick McKenna . <i>Synthetic Clarification and Correction Dialogues about Data-Centric Tasks: A Teacher-Student Approach</i> .	↪ Link
NeurIPS 2025	Jinyang Li, Jack Williams, Nick McKenna , Arian Askari, Nicholas Wilson, Reynold Cheng. <i>Unlocking SLM Potential for Data Analysis Code Generation via Non-Parametric Knowledge Distillation</i> .	↪ Link
AAAI 2025	Bhuvanashree Murugadoss, Christian Poelitz, Ian Drosos, Vu Le, Nick McKenna , Carina Suzana Negreanu, Chris Parnin, and Advait Sarkar. <i>Evaluating the Evaluator: Measuring LLMs’ Adherence to Task Evaluation Instructions</i> .	↪ Link
EMNLP Findings 2023	Nick McKenna* and Tianyi Li*; Liang Cheng, Mohammad Javad Hosseini, Mark Johnson, and Mark Steedman. <i>Sources of Hallucination by Large Language Models on Inference Tasks</i> .	↪ Link
AACL 2023	Nick McKenna , Tianyi Li, Mark Johnson, and Mark Steedman. <i>Smoothing Entailment Graphs with Language Models</i> . * Best Paper Award *	↪ Link
SustainNLP 2023	Nick McKenna and Priyanka Sen. <i>KGQA Without Retraining</i> . Workshop at ACL.	↪ Link
EMNLP 2021	Nick McKenna , Liane Guillou, Mohammad Javad Hosseini, Sander Bijl de Vroe, Mark Johnson, and Mark Steedman. <i>Multivalent Entailment Graphs for Question Answering</i> .	↪ Link
CASE 2021	Sander Bijl de Vroe* and Liane Guillou*; Miloš Stanojević, Nick McKenna , and Mark Steedman. <i>Modality and Negation in Event Extraction</i> . Workshop at ACL.	↪ Link
*SEM 2020	Nick McKenna and Mark Steedman. <i>Learning Negation Scope from Syntactic Structure</i> .	↪ Link